
2019-Ph-H1-Q2

Section: Our Dynamic Universe

Topic: Energy and Power

Question:

A skier of mass 70 kg descends a slope of vertical height 40 m. Assuming no friction, what is the skier's speed at the bottom of the slope?

Answer:

Using conservation of energy:

Gravitational potential energy lost = kinetic energy gained

$$mgh = \frac{1}{2}mv^2$$

$$70 \times 9.8 \times 40 = \frac{1}{2} \times 70 \times v^2$$

Mass cancels:

$$9.8 \times 40 = \frac{1}{2}v^2$$

$$392 = \frac{1}{2}v^2 \Rightarrow v^2 = 784 \Rightarrow v = 28 \text{ m/s}$$

Correct Option: D

Guidance for Students:

Energy conservation is a powerful tool: if no energy is lost to friction, potential energy simply converts into kinetic.

Revision Tips:

- Use $mgh = \frac{1}{2}mv^2$ when energy is conserved.
 - Mass cancels if no friction or external forces act.
 - Don't forget to **take the square root** when solving for speed.
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